

Chapter 2

PRAM: Matrix

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Matrix addition

A[nxn]

a_{11}	a_{12}	...	a_{1n}
a_{21}	a_{22}	...	a_{2n}
...	...	...	...
a_{n1}	a_{n2}	...	a_{nn}

+

B[nxn]

b_{11}	b_{12}	...	b_{1n}
b_{21}	b_{22}	...	b_{2n}
...	...	...	...
b_{n1}	b_{n2}	...	b_{nn}

=

C[nxn]

c_{11}	c_{12}	...	c_{1n}
c_{21}	c_{22}	...	c_{2n}
...	...	...	...
c_{n1}	c_{n2}	...	c_{nn}

C[nxn]

$c_{11} = a_{11} + b_{11}$	$c_{12} = a_{12} + b_{12}$	...	$c_{1n} = a_{1n} + b_{1n}$
$c_{21} = a_{21} + b_{21}$	$c_{22} = a_{22} + b_{22}$	...	$c_{2n} = a_{2n} + b_{2n}$
...	...	...	...
$c_{n1} = a_{n1} + b_{n1}$	$c_{n2} = a_{n2} + b_{n2}$	...	$c_{nn} = a_{nn} + b_{nn}$



Matrix addition: PRAM with $n \times n$ processors

$C[n \times n]$

$c_{11} = a_{11} + b_{11}$	$c_{12} = a_{12} + b_{12}$	...	$c_{1n} = a_{1n} + b_{1n}$
P_{11}	P_{12}		P_{1n}
$c_{21} = a_{21} + b_{21}$	$c_{22} = a_{22} + b_{22}$	...	$c_{2n} = a_{2n} + b_{2n}$
P_{21}	P_{22}		P_{2n}
...	...	...	...
...	...	...	...
$c_{n1} = a_{n1} + b_{n1}$	$c_{n2} = a_{n2} + b_{n2}$	...	$c_{nn} = a_{nn} + b_{nn}$
P_{n1}	P_{n2}		P_{nn}

➤ All ($n \times n$) processors run '+' in parallel: $O(1)$ =>

▪ P_{ij} processor:

- Read: a_{ij} & b_{ij}
- Write: c_{ij}

No overlapping data

- CRCW: $O(1)$
- EREW: $O(1)$



Matrix addition: PRAM with n processors

➤ All n processors run '+' in parallel in n steps: $O(n)$??? =>

$C[n \times n]$

Step 1	Step 2	...	Step n
$c_{11} = a_{11} + b_{11}$	$c_{12} = a_{12} + b_{12}$	...	$c_{1n} = a_{1n} + b_{1n}$
P_1	P_1	...	P_1
$c_{21} = a_{21} + b_{21}$	$c_{22} = a_{22} + b_{22}$	...	$c_{2n} = a_{2n} + b_{2n}$
P_2	P_2	...	P_2
...	...	...	...
...	...	...	...
$c_{n1} = a_{n1} + b_{n1}$	$c_{n2} = a_{n2} + b_{n2}$	...	$c_{nn} = a_{nn} + b_{nn}$
P_n	P_n	...	P_n

▪ P_{ij} processor:

○ Read: a_{ij} & b_{ij}

○ Write: c_{ij}

No overlapping data

○ CRCW: $O(n)$

○ EREW: $O(n)$

Your algorithm with
 k processors ($k \ll n$)?



Matrix multiplication

$$C_{ij} = a_{i1} * b_{1j} + a_{i2} * b_{2j} + \dots + a_{in} * b_{nj}$$

A[nxn]

a_{11}	a_{12}	...	...	a_{1n}
...	...	...	...	...
a_{i1}	a_{i2}	...	...	a_{in}
...	...	...	...	...
a_{n1}	a_{n2}	...	...	a_{nn}

B[nx n]

b_{11}	...	b_{1j}	...	b_{1n}
...	...	b_{2j}	...	...
...	...	...	...	...
...	...	...	...	...
b_{n1}	...	b_{nj}	...	b_{nn}

C[nxn]

c_{11}	...		...	c_{1n}
...	...		...	...
		c_{ij}		
...	...	...	...	...
c_{n1}	...	c_{nj}	...	c_{nn}

 C_{ij}

- $C_{ij} = a_{i1} * b_{1j} + a_{i2} * b_{2j} + \dots + a_{in} * b_{nj}$
- Vector: $A[i^{\text{th}} \text{ row}] \times B[j^{\text{th}} \text{ column}]$
- Number of operations = $\underbrace{n * \text{opt}(' * ')}_{O(1)} + \underbrace{(n-1) * \text{opt}(' + ')}_{O(\log(n))}$
- n processors $\sim$
 $\sim O(\log(n))$

 $C[n \times n]$

C_{11}	...		...	C_{1n}
...	...		...	...
		C_{ij}		
...	...	...	...	...
C_{n1}	...	C_{nj}	...	C_{nn}



PRAM

Matrix multiplication

- C_{ij} using n processors: $O(\log(n))$
- $C[n \times n]$: $(n*n)C_{ij}$
 - n processors: $O(n^2 * \log(n))$
 - n^3 processors: $O(\log(n))$
 - n^2 processors???

$C[n \times n]$

C_{11}	...		...	C_{1n}
...	...		...	...
		C_{ij}		
...	...	...	...	...
C_{n1}	...	C_{nj}	...	C_{nn}



PRAM

- C_{ij} using n processors: $O(\log(n))$
- $C[n \times n]$: $(n \times n)C_{ij}$
 - n processors: $O(n^2 * \log(n))$
 - n^3 processors: $O(\log(n))$
 - n^2 processors???

=> Difference between **CRCW** & **EREW**???

$C[n \times n]$

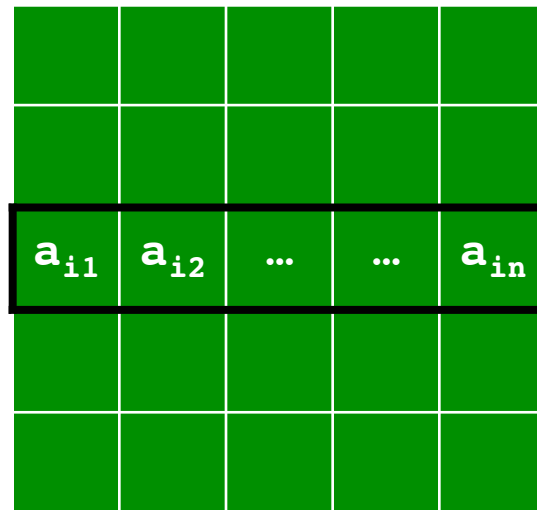
C_{11}	...		...	C_{1n}
...	...		...	...
		C_{ij}		
...	...	...	...	...
C_{n1}	...	C_{nj}	...	C_{nn}



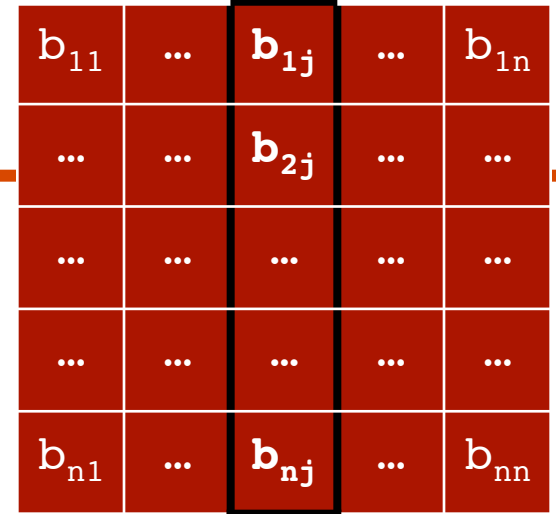
Row

$$\begin{aligned} C_{i1} &= a_{i1} * b_{11} + a_{i2} * b_{21} + \dots + a_{in} * b_{n1} \\ C_{i2} &= a_{i1} * b_{12} + a_{i2} * b_{22} + \dots + a_{in} * b_{n2} \\ &\dots \\ C_{in} &= a_{i1} * b_{1n} + a_{i2} * b_{2n} + \dots + a_{in} * b_{nn} \end{aligned}$$

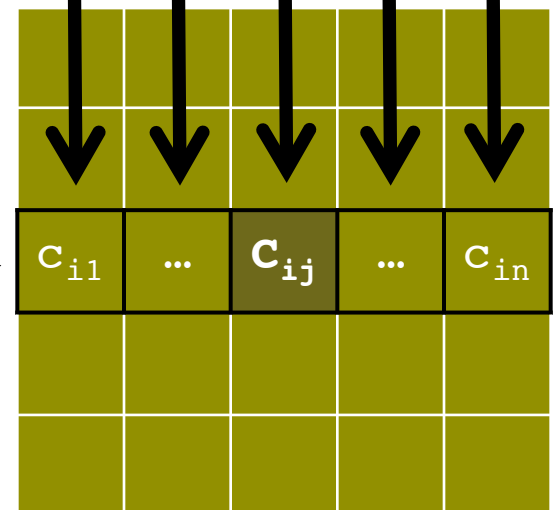
A[nxn]



B[nxn]



C[nxn]



Concurrent Read: in row

$$\begin{array}{l}
 C_{i1} = a_{i1} * b_{11} + a_{i2} * b_{21} + \dots + a_{in} * b_{n1} \\
 C_{i2} = a_{i1} * b_{12} + a_{i2} * b_{22} + \dots + a_{in} * b_{n2} \\
 \dots \\
 C_{in} = a_{i1} * b_{1n} + a_{i2} * b_{2n} + \dots + a_{in} * b_{nn}
 \end{array}$$

$P_1 \dots P_n$
 In parallel

CR: working
 ER: problem

CW: working
 EW: working



Column

$$\begin{aligned} C_{1j} &= a_{11} * b_{1j} + a_{12} * b_{2j} + \dots + a_{1n} * b_{nj} \\ C_{2j} &= a_{21} * b_{1j} + a_{22} * b_{2j} + \dots + a_{2n} * b_{nj} \\ &\dots \\ C_{nj} &= a_{n1} * b_{1j} + a_{n2} * b_{2j} + \dots + a_{nn} * b_{nj} \end{aligned}$$

A[nxn]

a_{11}	a_{12}	...	...	a_{1n}
...	...	...	...	...
a_{i1}	a_{i2}	...	...	a_{in}
...	...	...	...	...
a_{n1}	a_{n2}	...	...	a_{nn}

B[nxn]

b_{11}	...	b_{1j}	...	b_{1n}
...	...	b_{2j}	...	...
...	...	...	...	...
...	...	...	...	...
b_{n1}	...	b_{nj}	...	b_{nn}



C[nxn]

		C_{1j}		
		...		
		C_{ij}		
		...		
		C_{nj}		



Column

$$\begin{array}{l} C_{1j} = a_{11} * b_{1j} + a_{12} * b_{2j} + \dots + a_{1n} * b_{nj} \\ C_{2j} = a_{21} * b_{1j} + a_{22} * b_{2j} + \dots + a_{2n} * b_{nj} \\ \dots \\ C_{nj} = a_{n1} * b_{1j} + a_{n2} * b_{2j} + \dots + a_{nn} * b_{nj} \end{array}$$

$P_1 \dots P_n$
In parallel

CR: working
ER: problem

CW: working
EW: working



PRAM EREW with n^2 processors

a_{11}	a_{12}	...	a_{1n}			
a_{21}	a_{22}	...	a_{2n}			
...	...	...	...	...	...	
a_{n1}	...	a_{nn}	a_{n1}	a_{n2}	...	a_{nn}



P_{11}	P_{12}	...	P_{1n}
P_{21}	P_{22}	...	P_{2n}
...	...	...	...
P_{n1}	P_{n2}	...	P_{nn}

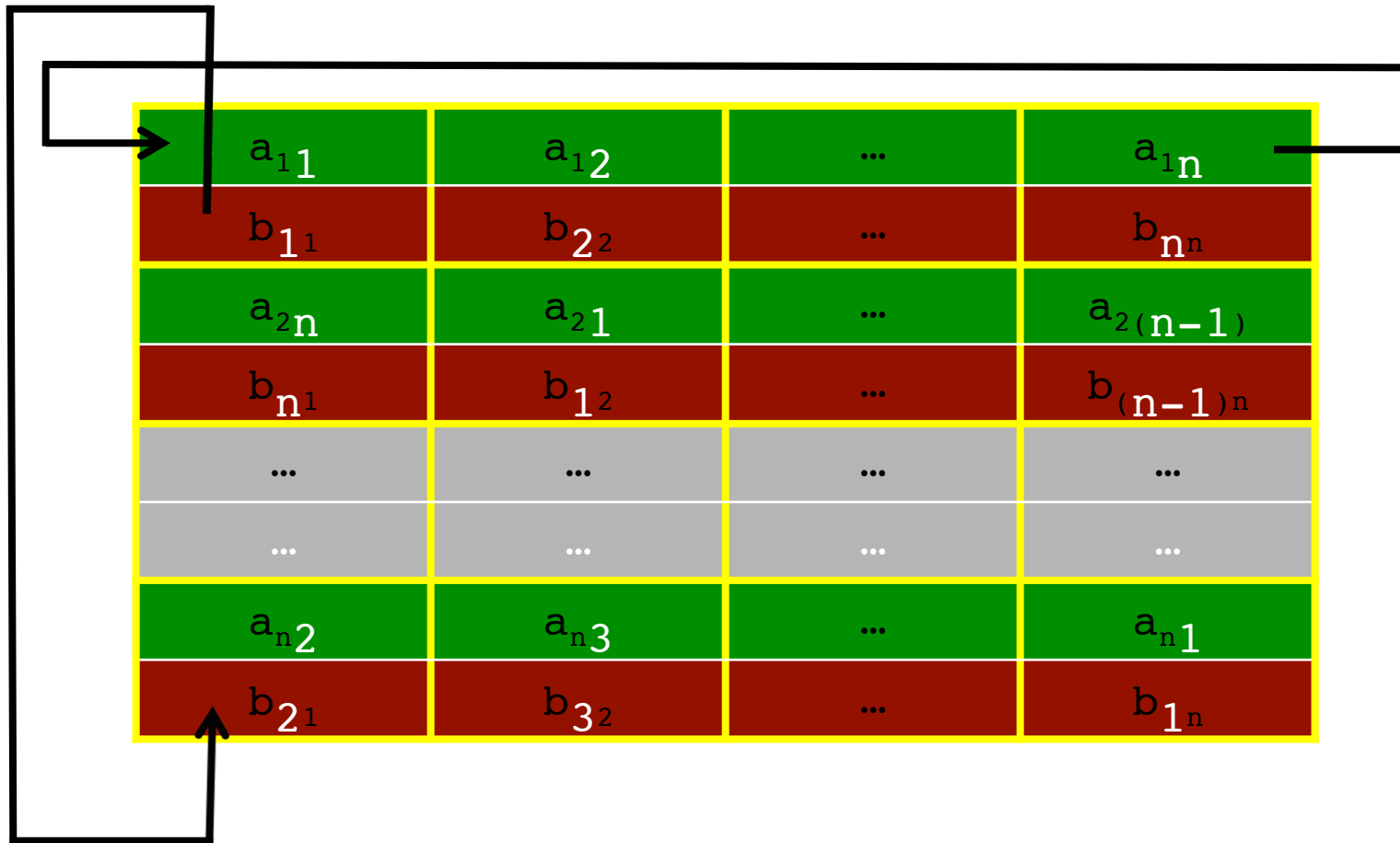
Why not use $n^3 > n^2$ processors with EREW?

b_{n1}			
$b_{(n-1)1}$	...		
...	...	$b_{n(n-1)}$	
b_{11}	...	...	b_{nn}
b_{n1}	...	$b_{2(n-1)}$	...
...	...	$b_{1(n-1)}$	b_{2n}
b_{21}	...	$b_{n(n-1)}$	b_{1n}



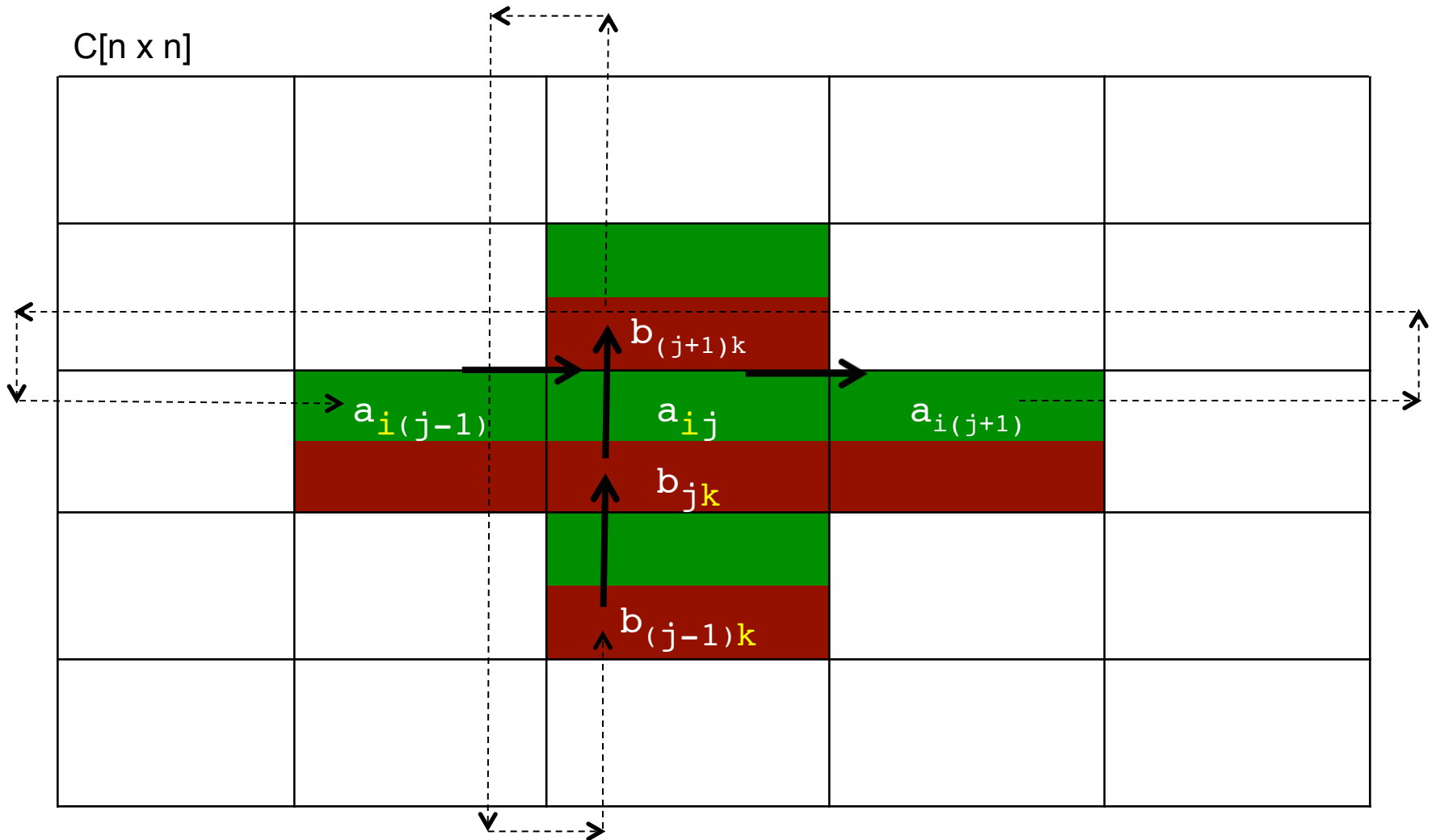


PRAM EREW with n^2 processors





PRAM EREW with n^2 processors





PRAM EREW with n^2 processors

a_{11}	a_{12}	...	a_{1n}			
a_{2n}	a_{21}	a_{22}	...	a_{2n}		
...	...	...	...	...	...	
a_{n2}	...	a_{nn}	a_{n1}	a_{n2}	...	a_{nn}



P_{11}	P_{12}	...	P_{1n}
P_{21}	P_{22}	...	P_{2n}
...	...	...	...
P_{n1}	P_{n2}	...	P_{nn}

- Each P_{ij}
 - Run "*" in n steps sequentially
 - Run "+" in n steps
- $O(n)$



b_{n1}			
$b_{(n-1)1}$	...		
...	...	$b_{n(n-1)}$	
b_{11}	...	...	b_{nn}
b_{n1}	...	$b_{2(n-1)}$	...
...	...	$b_{1(n-1)}$	b_{2n}
b_{21}	...	$b_{n(n-1)}$	b_{1n}



PRAM with n processors

$$C_{i1} = a_{i1} * b_{11} + a_{i2} * b_{21} + \dots + a_{in} * b_{n1}$$

...

$$C_{ij} = a_{i1} * b_{1j} + a_{i2} * b_{2j} + \dots + a_{in} * b_{nj}$$

...

$$C_{in} = a_{i1} * b_{1n} + a_{i2} * b_{2n} + \dots + a_{in} * b_{nn}$$

$A[n \times n]$

P_i : Row $A[i]$

P_i : Column $B[j]$

P_i : $B[n \times n]$

a_{i1}	a_{i2}	...	...	a_{in}

$B[n \times n]$

b_{11}	...	b_{1j}	...	b_{1n}
...	...	b_{2j}	...	...
...	...	...	...	...
...	...	...	...	...
b_{n1}	...	b_{nj}	...	b_{nn}

$C[n \times n]$

					P_1
					...
C_{i1}	...	C_{ij}	...	C_{in}	P_i
					...
					P_n



PRAM

CRCW with n processors

Each P_i :

- $C_{ij} = a_{i1} * b_{1j} + a_{i2} * b_{2j} + \dots + a_{in} * b_{nj} \Rightarrow O(n)$
- Row $C[i]: O(n^2)$



PRAM

EREW with n processors

$$C_{i1} = a_{i1} * b_{11} + a_{i2} * b_{21} + \dots + a_{in} * b_{n1}$$

...

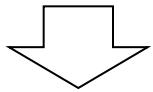
$$C_{ij} = a_{i1} * b_{1j} + a_{i2} * b_{2j} + \dots + a_{in} * b_{nj}$$

...

$$C_{in} = a_{i1} * b_{1n} + a_{i2} * b_{2n} + \dots + a_{in} * b_{nn}$$

P_i : Row $A[i]$

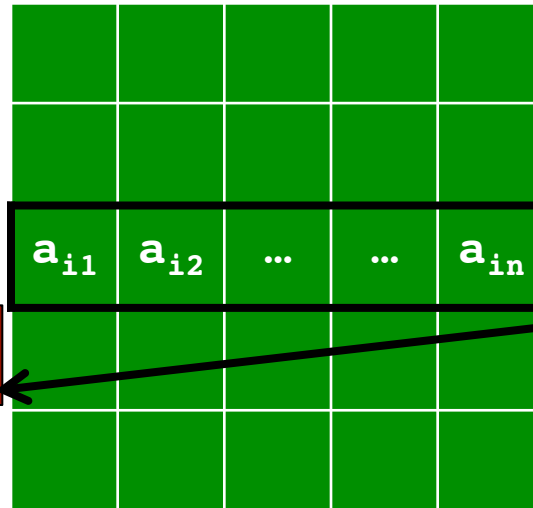
P_i : Column $B[j]$



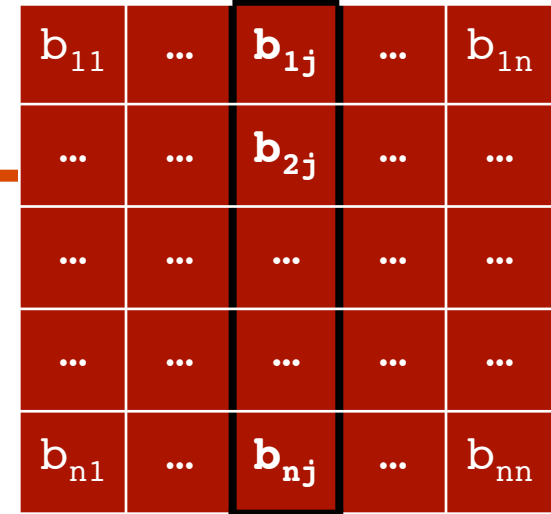
Each P_i : $B[n \times n]$

ER???

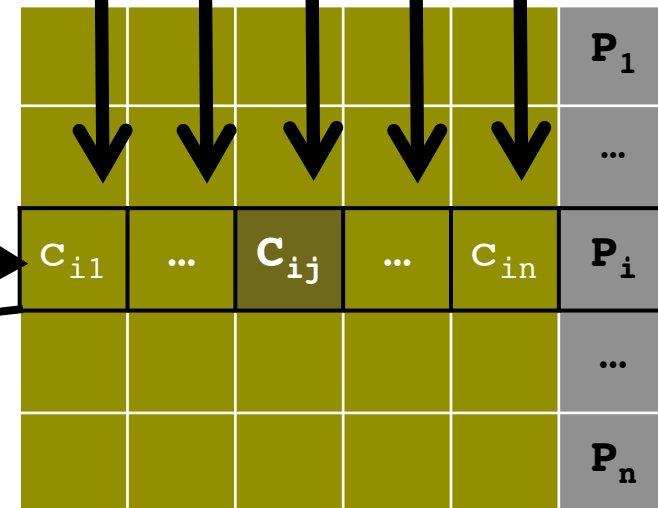
$A[n \times n]$



$B[n \times n]$



$C[n \times n]$



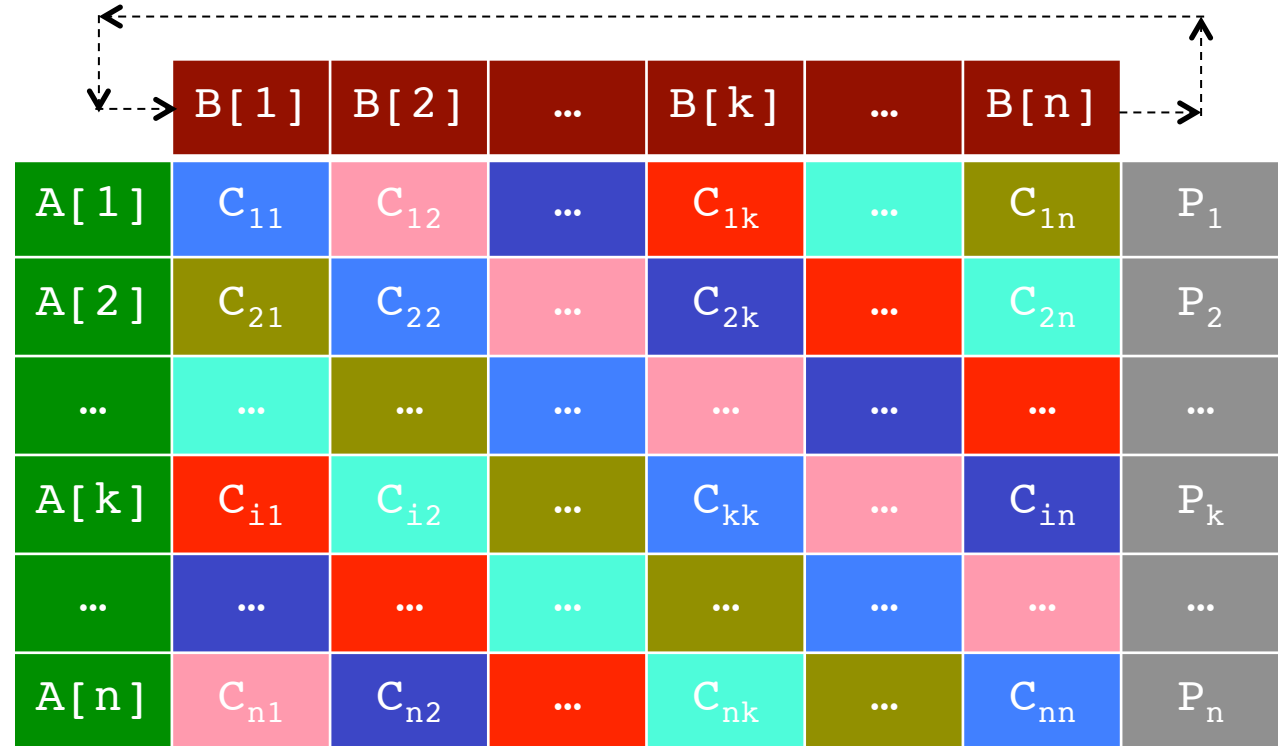


PRAM

EREW with n processors

Each P_i :

- $C_{ij} = a_{i1} * b_{1j} + a_{i2} * b_{2j} + \dots + a_{in} * b_{nj} \Rightarrow O(\mathbf{n})$
- Row $C[i]: O(\mathbf{n}^2)$





PRAM

EREW with k processors ($k \ll n$)

Each P_i :

- 1 step: $O(k \cdot n) \Rightarrow$
- 1 phase (n/k steps): $O(k \cdot n \cdot n/k) \sim O(n^2)$
- n/k phases: $O(n^2 \cdot n/k) \sim O(n^3/k)$

